



The 5 Z's that close out **A → Z** for AI engineers.

Zero-shot prompting, Z-score, ZeRO and more. **Plain-English** definitions, one swipe each, plus four Z-tools shaping the modern AI stack.

ZERO-SHOT LEARNING • LEARNING

Zero-shot Learning

unseen • classes • still recognized

Show a model categories it has never seen, and zero-shot learning still sorts them out. It reasons from **shared traits and descriptions** instead of labelled examples, so one model can stretch to brand-new tasks.

USED IN

- Vision & NLP
- CLIP-style models
- Generalization



SEEN → UNSEEN VIA ATTRIBUTES

ZERO-SHOT PROMPTING • INFERENCE

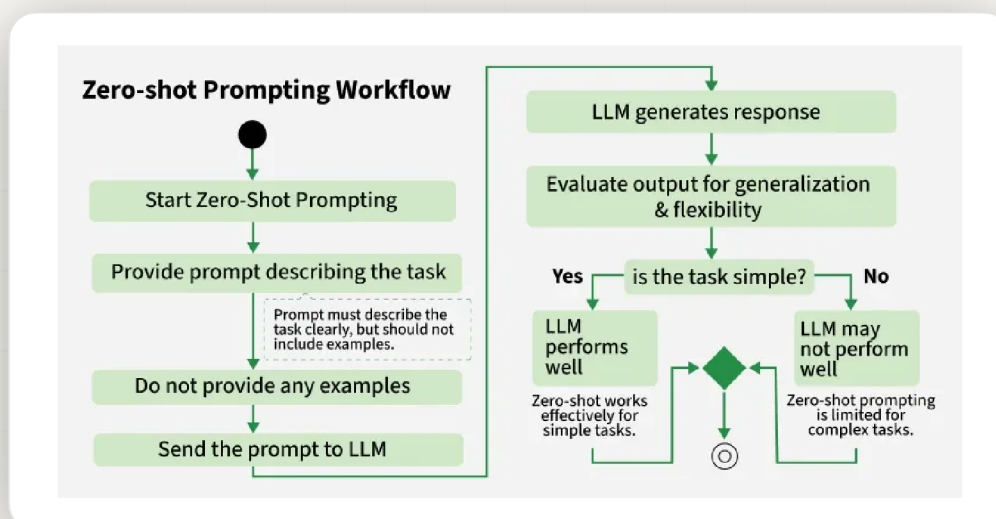
Zero-shot Prompting

just • ask • no examples

No examples, no warm-up. You hand over an instruction and the model answers straight from **what it already knows**. It is the first thing to try, and the baseline every fancier prompting trick has to beat.

USED IN

- Prompt engineering
- LLM apps
- Quick tasks



INSTRUCTION-ONLY PROMPT

ZERO-SHOT CODING • AGENTS

Zero-shot Coding

prompt • → • working code

Describe the feature in plain English and the model writes the code, with **no sample snippets or scaffolding** to copy. This is the loop powering Cursor, Claude Code, and the whole vibe-coding wave.

USED IN

• Coding agents

• Cursor / Claude Code

• Vibe coding

prompt
"build a CSV parser in TS"



output
parser.ts ✓

NO EXAMPLES NEEDED

Z-SCORE • STATISTICS

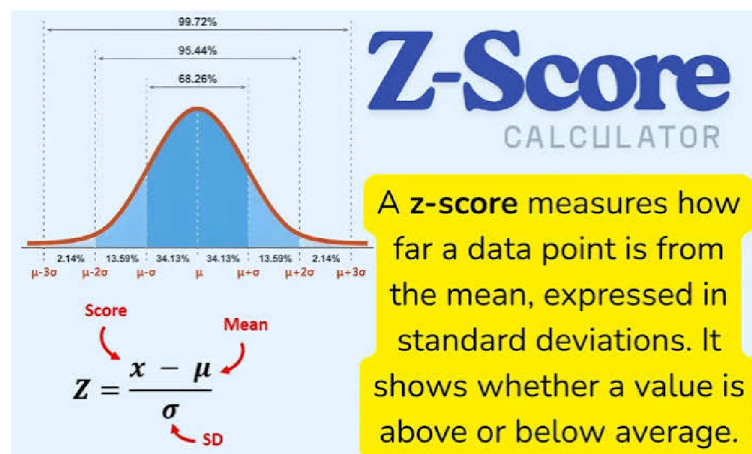
Z-score

$$(x - \mu) / \sigma$$

How far from normal is this number? A z-score answers in **standard deviations from the mean**. That one value drives outlier detection, normalization, and feature scaling across ML pipelines.

USED IN

- Outlier detection
- Normalization
- Feature scaling



DISTANCE FROM THE MEAN

ZERO • INFRASTRUCTURE

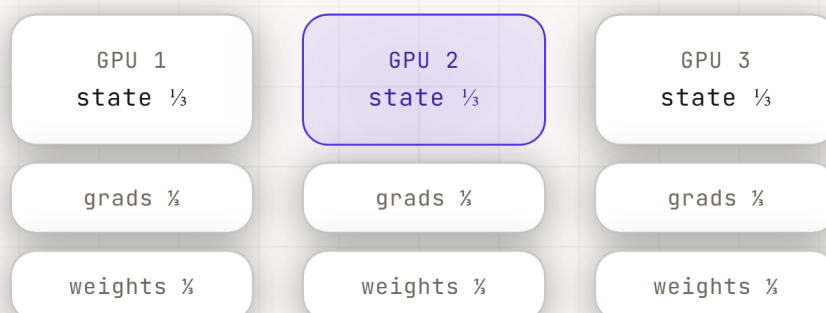
ZeRO

zero redundancy • optimizer

Big models run out of GPU memory fast. ZeRO, the Zero Redundancy Optimizer in DeepSpeed, **shards optimizer state, gradients, and weights across GPUs** instead of copying them everywhere, so billion-parameter training suddenly fits.

USED IN

- DeepSpeed
- LLM training
- Multi-GPU



NO DUPLICATED STATE

↓ UP NEXT • TOOLS

Tools next.

Beyond the **5 Z-words** sit four tools shaping different corners of the modern AI stack.



INDIAN SAAS SUITE

Zoho

CRM • email • accounting • 50+ apps



India's privacy-first software house, quietly running a **50-app empire** across CRM, mail, finance, and HR. Its Zia AI layer now threads through the whole suite.

WHAT IT IS

Cloud business-app suite from India.

IN AI STACKS

Zia AI across CRM, analytics, support.

WHY TEAMS USE IT

50+ integrated apps, privacy-first.

PAIRS WITH

Your existing biz workflows.

HIGH-PERFORMANCE CODE EDITOR

Zed



Rust-built • multiplayer • AI-native

Built in Rust by the team behind Atom, Zed feels instant. **Native multiplayer and built-in AI** come standard, tuned for how developers actually code alongside an LLM.

WHAT IT IS

Rust-built collaborative code editor.

IN AI STACKS

In-editor chat, agents, pair-coding.

WHY DEVS USE IT

Fast, native, AI & multiplayer baked-in.

PAIRS WITH

Claude, GPT, Ollama, your own keys.

NO-CODE AUTOMATION

Zapier

7,000+ apps • workflows • AI agents



The original duct tape of the web. Zapier wires **7,000+ apps into automated workflows**, now with AI steps, agents, and chatbots, still no engineer required.

WHAT IT IS

No-code automation across SaaS apps.

IN AI STACKS

Agents, chatbots, AI-in-the-loop ops.

WHY TEAMS USE IT

Build AI workflows without writing code.

PAIRS WITH

OpenAI, Claude, your CRM & tools.

CHINESE FRONTIER LAB

Zhipu AI

Z.ai • GLM, ChatGLM, GLM-4.5



A Tsinghua spinout, globally rebranded as Z.ai, building the **open-weight GLM family on Chinese silicon**. It keeps hitting frontier benchmarks while staying open.

WHAT IT IS

China's leading open-model frontier lab.

KNOWN FOR

GLM, ChatGLM, GLM-4.5, AI agents.

WHY IT MATTERS

SOTA open weights, trained on local chips.

PAIRS WITH

HF, vLLM, your own self-host stack.

The **Z** stack, and the **A→Z**, done.

Save the full set. Thanks for swiping along. The whole alphabet now lives in your feed.

01 **Zero-shot Learning** learning

02 **Zero-shot Prompting** inference

03 **Zero-shot Coding** agents

04 **Z-score** statistics

05 **ZeRO** infrastructure

+ **Zoho** tool

+ **Zed** tool

+ **Zapier** tool

+ **Zhipu AI** tool



FOLLOW FOR THE FULL A TO Z

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★ Done